

5. VIBROTOME

Preparation of relatively thick sections from unfixed or fixed tissue using a vibrating blade, commonly for delicate tissue and three-dimensional studies.

SOP No.: LAB-HISTO-05	Revision: 01	Effective: _____
Department: Histopathology / Research	Prepared by: _____	Approved by: _____

Purpose

- To cut tissue sections with reduced compression and distortion compared with conventional microtomy, particularly for brain, nerve and other delicate specimens.

PPE & Safety

- Wear lab coat and gloves; use eye protection where splashing is possible. Handle the vibrating blade as a sharp instrument. Secure the specimen and keep hands away from the blade during operation. Treat tissue and cutting fluids as potentially biohazardous.

Materials

- Fixed or prepared tissue; specimen tray; appropriate buffer/cutting medium; vibrating blade; specimen adhesive/holder as applicable; labelled slides or collection wells; forceps.

Procedure

- Inspect the blade, tray, specimen holder and vibration mechanism. Select the validated cutting speed, amplitude and section thickness for the tissue.
- Mount the specimen securely and immerse it in the appropriate chilled/room-temperature buffer as specified by the laboratory method.
- Set blade position and clearance with the instrument stopped/secured.
- Start the instrument and advance the blade smoothly through the tissue. Avoid excessive feed force or repeated passes.
- Collect sections gently in buffer or onto labelled slides, according to the downstream staining/imaging protocol.
- Stop the instrument before removing the specimen or blade. Clean and disinfect the tray and work surfaces.

Quality Control

- Sections should have consistent thickness and minimal tearing, compression or vibration artefact. Check orientation and tissue integrity before downstream processing.

Troubleshooting

- Tearing → check blade sharpness and tissue fixation/softness. Compression → reduce cutting stress and verify vibration settings. Uneven thickness → check specimen fixation and holder stability.

Shutdown & Cleaning

- Turn off vibration and secure the blade before cleaning. Dispose of contaminated fluids according to laboratory waste procedures.

Important: This is a general SOP template. Before operational use, insert the exact manufacturer model, validated temperatures/times, reagent names, decontamination procedure, QC acceptance criteria and institutional biosafety/chemical-safety requirements.